



Failure modes and energy evolution of basalt in uniaxial compression tests after cryogenic freeze-thaw cycles

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Abstract

With the gradual deepening of human extraterrestrial exploration, the extreme temperatures in the environment of celestial bodies have become key research focuses. In environments of celestial bodies such as the Moon, the huge difference and frequent change in the surface temperature have exposed rocks to the freeze-thaw (F-T) cycles. Meanwhile, F-T damage is also a non-negligible natural disaster in cold-region engineering. Uniaxial compression mechanical tests were conducted on basalt specimens subjected to cryogenic F-T cycles ($-80\text{ }^{\circ}\text{C} \sim 100\text{ }^{\circ}\text{C}$), during which acoustic emission (AE) monitoring was used to record the damage evolution process in the specimens. Test results show that after experiencing multiple F-T cycles, the P-wave velocity of basalt specimens decreases by 12%, the uniaxial compressive strength declines by 26% ~ 33%, and the volume with F-T damage accounts for 31.6% ~ 46.7% of the total. In addition, for basalt specimens subjected to 15 and 25 F-T cycles, the damage areas show radial distances of 7.9 mm and 11.7 mm from the surface. The proportion of tensile failure in the specimens increases from 10.3 to 28.8% (25 cycles). The scanning electron microscopy (SEM) was utilized for microscopic analysis of fracture planes of the tested basalt. New fractures showing the characteristics of intergranular failure are observed on the fracture planes of specimens undergoing F-T cycling, which causes more tensile failure in the specimens. The test results provide not only basic data for research into mechanical behaviors of rocks with F-T damage but also offer theoretical guidance for building construction and slope design and construction in alpine regions.

Keywords Freeze-thaw cycles · Basalt · Scanning electron microscopy · Acoustic emission

Introduction

With the gradual depletion of many resources on Earth, harvesting resources from objects in the outer space including asteroids, the Moon, and Mars to replenish dwindling reserves of resources on Earth will become an inevitable

trend in the future (Dallas et al. 2020). However, the surface temperature of objects in the outer space shows extreme fluctuations. For example, the annual mean temperature of Mars is about $-63\text{ }^{\circ}\text{C}$ (Xu et al. 2023); due to solar radiation, the Moon exhibits a huge difference in extreme temperatures in local regions (Williams et al. 2019). The complex freeze-thaw (F-T) cycles are a key factor causing erosion and deterioration of bare rocks and very likely to induce engineering geological disasters. For instance, rock slopes in alpine regions may collapse and become unstable under F-T cycles, and tunnel entrances in cold regions may collapse (Huang et al. 2022; Hewage et al. 2023; Lu et al. 2023; Zhu et al. 2024). Therefore, rock mass instability induced by frequent F-T cycles is not only an engineering problem facing those working in geotechnical engineering but also a research hot-spot for those interested in rock mechanics. The strength of rocks under F-T cycles is dependent on a series of complex characteristics of rocks, including the mineral compositions, pore size and distribution, permeability, and mechanical

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behaviors of rocks (De Kock et al. 2015; Kim et al. 2021; Chen et al. 2022; Liu et al. 2022). Multiple discontinuities are present in natural rocks, such as pores, cracks, joints, cracked weak planes, and faults (Bobet and Einstein 1998; Wong and Einstein 2009). When the temperature in an alpine region is zero (plus or minus variations induced in the freezing point of water by the prevailing atmospheric pressure), pore water in the rocks freezes to ice, the volume of which expands by 9%, thus elevating the pressure in the pores and micro-fractures. This, in turn, enlarges existing pores and cracks, and induces new cracks in the rocks (Freire-Lista et al. 2015; Song et al. 2021; Fan et al. 2022).

At present, research focuses on the influence of F-T cycles on different rocks. İnce and Fener investigated the influences of F-T cycles on the physico-mechanical behaviors of pyroclastic rocks (İnce and Fener 2016). Ghobadi et al. evaluated the influence of F-T damage on tuff and determined that F-T cycles impair the integrity of the rock (Ghobadi et al. 2016). Jiang quantitatively expounded the substantial influence of F-T cycles on the damage and deterioration of internal structures of andesite (Jiang 2018). Cao et al. investigated changes in the indices of P-wave velocity and mechanical behaviors of granite experiencing F-T cycles (Cao et al. 2024). Its high porosity and water absorbing capacity meant that sandstone has been widely studied under F-T cycles (Ren et al. 2022; Xiao et al. 2024). The above research has shown that different rocks show dissimilar changes under F-T cycles, while the characteristics of basalt under F-T conditions have not been fully explored. Basalt, as a dense rock, is formed by cooling and condensation of magma ejected from volcanos and has been widely applied to the construction and engineering sectors. Basalt, as one of the ideal materials for underground engineering construction, requires specific treatment and research due to its unique mechanical and structural characteristics (Zhou and Lei 2019; Asryan and Rots 2024). Therefore, domestic and foreign scholars have studied the mechanical behaviors of basalt in different regions. Jin et al. selected similar materials to simulate basalt with columnar joints and conducted uniaxial and triaxial compression tests as well as numerical simulation (Jin et al. 2018). Lu et al. investigated the influence of microwave treatment on mechanical behaviors of basalt under uniaxial compression and conventional triaxial compression (Lu et al. 2020). They found that the microwave irradiation time affects the mechanical behaviors of basalt, such as its strength and deformation. Latest research has shown that basalt, a rock formation widespread on the Moon and Mars (Li et al. 2022), can be probably used as a natural building material for constructing extraterrestrial bases on remote celestial bodies (De Hon 2022; Feng et al. 2024). Considering this, systematic investigation of the engineering properties of basalt under various

environmental conditions is required to establish a theoretical foundation for geotechnical applications in extreme temperature environments.

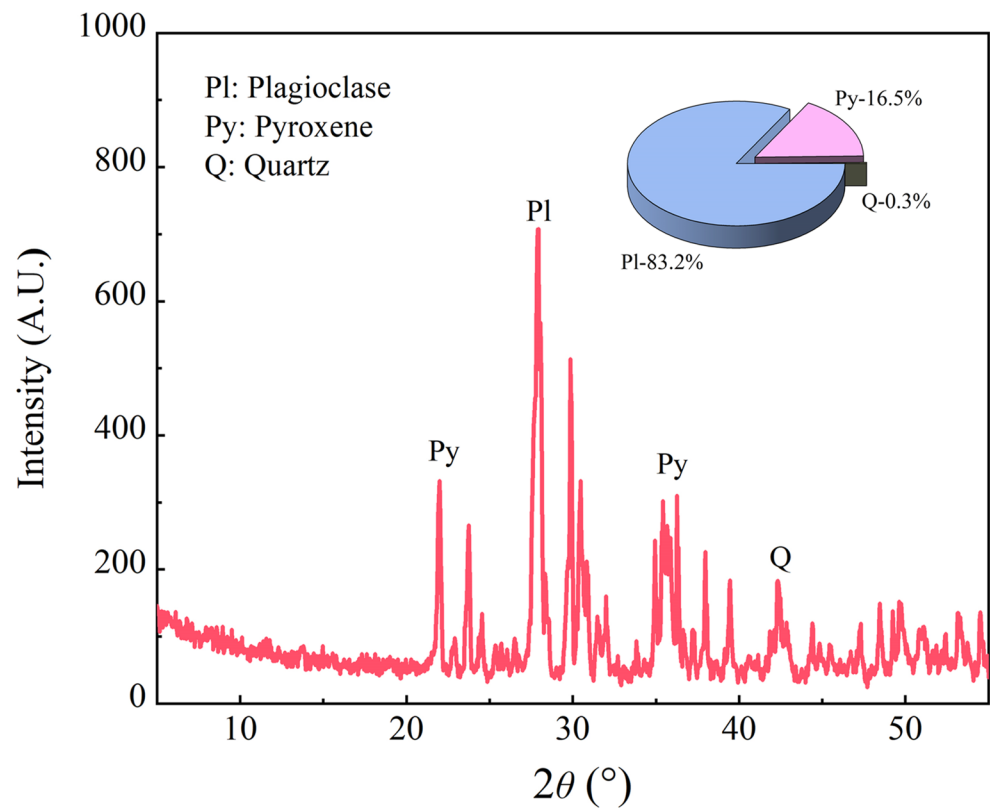
In existing research involving F-T cycling of rock, the limited range of temperature adjustment of most temperature boxes has become the key factor that restricts the broadening of such research. In previous research on the F-T cycles of rocks, the temperature can generally be adjusted in the range from $-40\text{ }^{\circ}\text{C}$ to $40\text{ }^{\circ}\text{C}$ (Tan et al. 2011; Liu et al. 2015; Huang et al. 2019; Deprez et al. 2020; Zhang et al. 2022; Mousavi and Rezaei 2022). There are few studies in which the low temperature limit can be lower than $-40\text{ }^{\circ}\text{C}$. In recent years, although the research direction of mechanical behaviors of frozen-thawed rocks has received much attention, research into changes in the characteristics of basalt in the cyclic cryogenic F-T process is scarce. Comprehensively understanding the mechanical behaviors of rocks in the cold regions to some extent is limited by the current state of research, which also puts cold-region engineering design and construction at potential risk.

In view of this, intact basalt specimens were selected as the research object; these were subjected to a cyclic F-T process with temperature varying from $-80\text{ }^{\circ}\text{C}$ to $100\text{ }^{\circ}\text{C}$. Thereafter, uniaxial compression tests were conducted on rock specimens and acoustic emission (AE) was adopted to monitor, in real-time, the whole process. Finally, scanning electron microscopy (SEM) was employed to observe the post-failure microstructures of rock specimens, allowing analysis and discussion of the failure modes. Aiming at the requirements for cold-region engineering practice, analysis of crack evolution and energy variation based on characteristics of rock specimens revealed in the F-T and compression process provides theoretical guidance for the design and construction of geotechnical engineering works in alpine regions.

Materials and methods

Specimen preparation

Considering the pervasiveness of rock in cold-region engineering and extraterrestrial planets, the rock selected here was basalt from a dormant volcano in China. Figure 1 shows the mineral compositions of basalt determined by X-ray diffraction (XRD). XRD analysis results reveal that the main minerals in the basalt were plagioclase and pyroxene, with a small amount of quartz filling interstices between mineral particles. The density and P-wave velocity of basalt specimens were 2.9 g/cm^3 and 5800 m/s , respectively. To reduce the influence of anisotropy on the test, basalt with the uniform appearance, good homogeneity, and appropriate block

Fig. 1 Compositional analysis of basalt

size was selected, cut, and polished. Through precision cutting and polishing, 30 standard cylindrical specimens ($\varnothing$ 50 mm $\times$ 100 mm) were initially prepared. From these, 12 specimens with closely matched P-wave velocities (variation $< 2\%$) were selected and equally allocated to three test groups for subsequent experimentation.

Test procedure

To simulate the temperature variation of rocks in the cold region, the upper and lower temperatures of F-T cycles were set to be -80°C and 100°C in the tests. Prior to freeze-thaw cycling, all basalt specimens were fully saturated through vacuum immersion in deionized water for 24 h until reaching mass stabilization, defined as less than 0.1% variation in 12-hour intervals. This saturation procedure followed established rock testing methodologies to ensure complete pore water saturation. Each F-T cycle was set to last 24 h. After being wrapped in plastic film, a basalt specimen was placed in an ultra-low-temperature refrigerator at -80°C for 12 h. Then, the frozen basalt specimen was placed in a high-temperature drying oven to be heated for 12 h. This protocol ensured complete temperature propagation and water-ice phase transition within the specimens. (Mousavi et al. 2019; Ren et al. 2022) Fig. 2 displays the specific procedure of the whole test. Figure 2(a) shows the setting of cryogenic F-T cycles in the tests; Fig. 2(b) shows the uniaxial compression

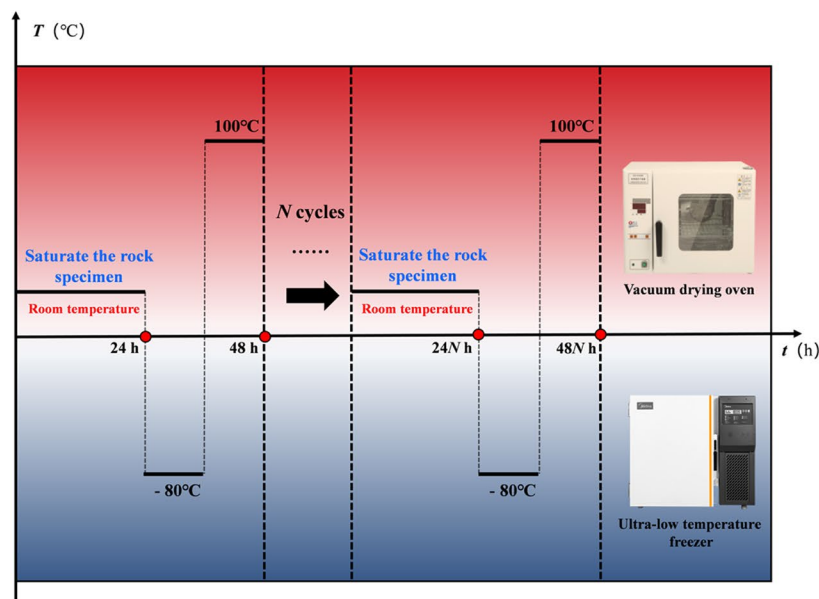
test and establishment of the AE monitoring system; Fig. 2(c) shows various components of the research scheme.

The numbers of F-T cycles were set to 0, 15, and 25 in the present research. The uniaxial compression tests were performed on a TAW-2000 microcomputer-controlled electro-hydraulic servo machine, of which the maximum bearing capacity was 2000 kN. It could simultaneously acquire data of the load, displacement, and time in the loading process, to attain the stress-strain curves and failure modes of basalt. The AE monitoring system was an SH-II multi-channel ultrasonic acquisition and processing system produced by Physical Acoustic Corporation, USA. The system mainly comprised of a PCI-2 host, Nano-30 probe sensors, and a 1220 A-AST preamplifier, of which the gain was 40 dB. AE software AEwin was employed to monitor (in real-time) and analyze parameters including the time of occurrence and energy of AE events.

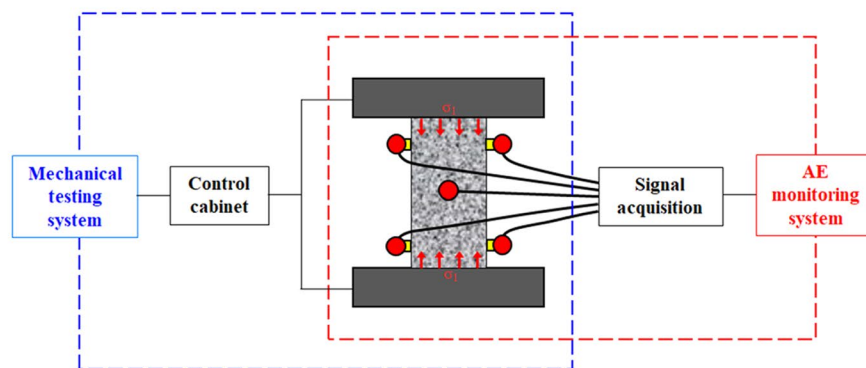
Experimental results

P-wave velocity

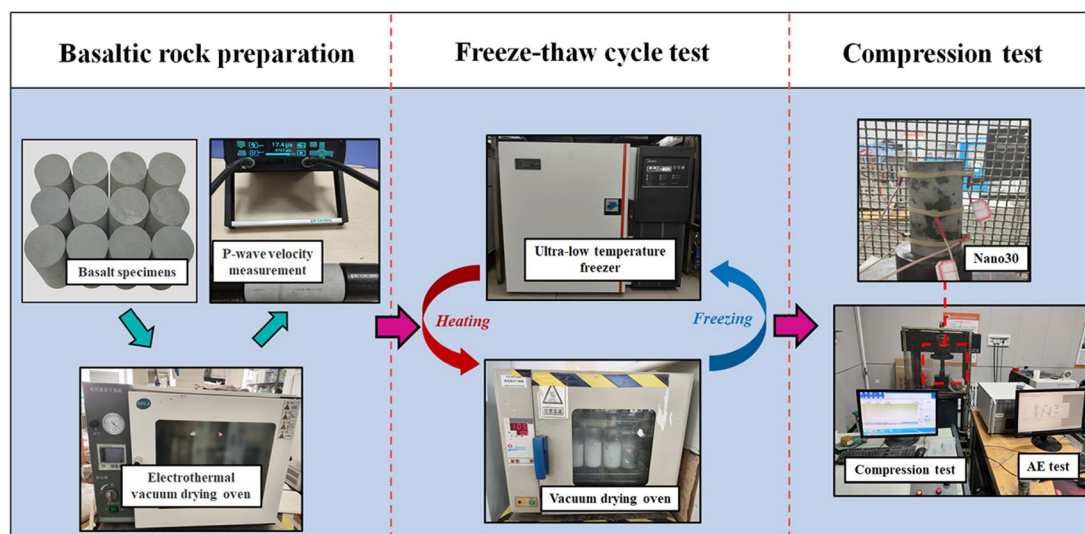
The P-wave velocity of rocks is mainly determined by the lithology, porosity, and clay content of rocks (Domenico 1985; Goldberg and Gurevich 1998). In the whole cyclic F-T test, the P-wave velocities of rocks experiencing



(a) Setting of cryogenic F-T cycles



(b) The compression test and establishment of the AE monitoring system



(c) Influence of cryogenic F-T cycles on the mechanical behaviors of basalt

◀ **Fig. 2** Specific procedure of each test

different numbers of F-T cycles were measured (Fig. 3). The P-wave velocity of basalt reduces with the increasing number of F-T cycles. After up to 25 F-T cycles, the P-wave velocity decreases by 12% from the initial 5834 m/s to 5113 m/s (on average). The foundational cause for the reduction of the P-wave velocity is that micro-fractures developed due to internal damage to rocks in the F-T environment alter the microstructures and mechanical behaviors. The trend is such that the P-wave velocity of the rock decreases with the increasing number of F-T cycles is consistent with the previous test results pertaining to sandstone, granite, and limestone (Liu et al. 2012; Eslami et al. 2018). This indicates that changes in the P-wave velocity of basalt under F-T cycles are universal.

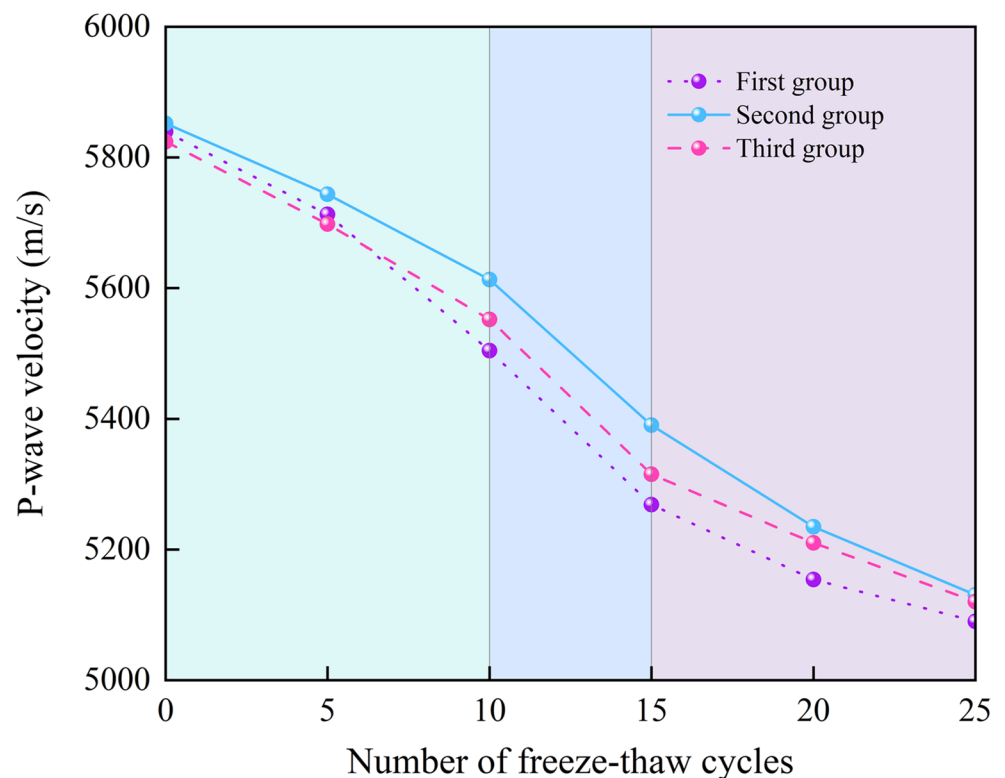
With the propagation of fracture networks, its influence on the P-wave velocity is probably no longer a simple Linear relationship. In the initial stage of F-T cycles when there are only a few micro-fractures, the P-wave velocity declines slowly and in a quasi-linear manner. After reaching 10 F-T cycles, the P-wave velocity attenuates more rapidly in subsequent F-T cycles due to the occurrence of new cracks and propagation of pre-existing cracks. After the fracture networks develop to a certain extent, the P-wave velocity decreases more slowly in the process from 15 to 25 F-T cycles. Such a relationship provides a new perspective for studying the long-term influence of F-T cycles on the

rock. The accurate measurement and analysis of variation of P-wave velocity allows a better understanding of the evolution of internal damage to rocks in the cyclic F-T process and provides a solid basis for the stability assessment and prediction of rocks in cold-region engineering.

Mechanical behaviors

Damage to rocks either due to F-T cycles or due to compression tests is, in essence, the initiation, development, and coalescence of micro-fractures in rocks. Damage to rocks is closely related to the activities of micro-fractures. Initiation of micro-fractures means the onset of initial damage in rocks, and the number, location, and strike of micro-fractures reflect the characteristics of initial damage (Fang et al. 2019). In the micro-fracture development stage, the rate of propagation, and interaction of micro-fractures determine the rate of development of damage to the rock. For example, if micro-fractures propagate fast and interweave with each other in a local area, the damage in the area is aggravated rapidly. When micro-fractures finally coalesce to form macro-cracks, the overall mechanical behaviors of rocks may change substantially and the bearing capacity decreases, suggesting that the rocks have been severely damaged. Under F-T cycles, microstructures in rock specimens also change correspondingly, the bonds between particles are inevitably damaged, microstructures become looser, and mechanical behaviors are deteriorated. With the

Fig. 3 Variation in the P-wave velocity of basalt after different numbers of F-T cycles



increasing number of F-T cycles, the degree of development of initial micro-fractures in rock specimens also increases.

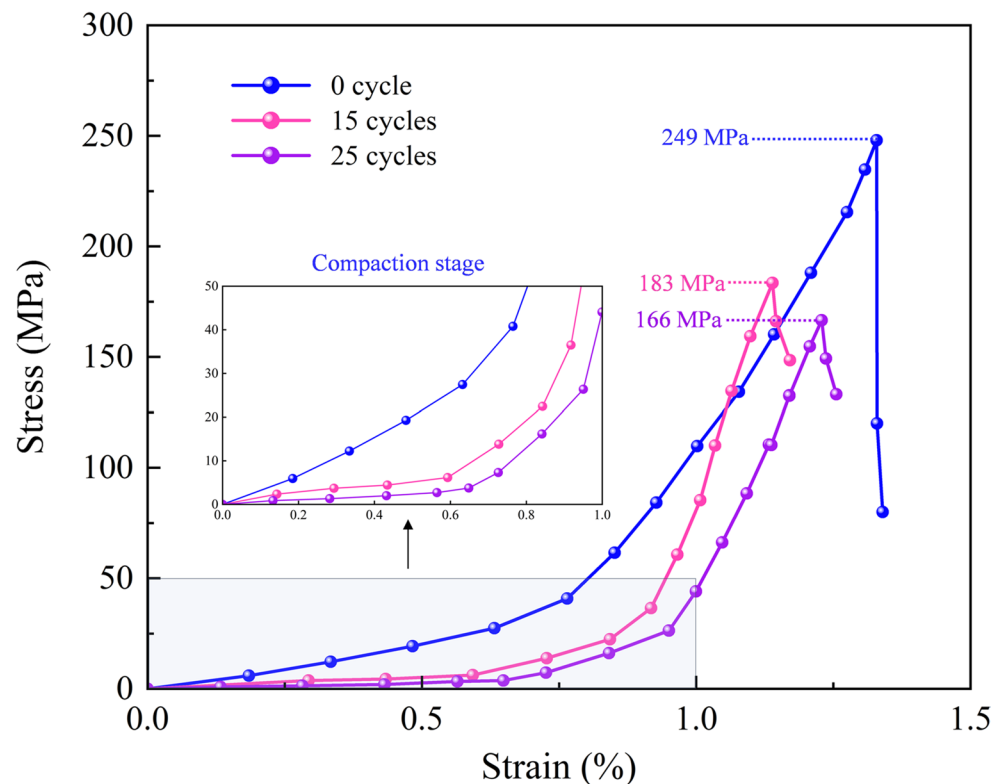
To explore the influences of F-T cycles and their number on mechanical behaviors of basalt, uniaxial compression tests, along with AE, were adopted to study the rock specimens. In this way, the failure modes and failure mechanisms of basalt subjected to F-T cycles were obtained.

Under the uniaxial compressive load, the whole process of deformation of rock specimens can be characterized by the stress–strain curves (Fig. 4). The stress–strain curves of rock specimens subjected to different numbers of F-T cycles in the uniaxial compression tests show that the rock specimens not subjected to F-T cycles show the highest uniaxial compressive strength. The uniaxial compressive strength gradually decreases with an increase in the number of F-T cycles. During the loading of rock specimens without F-T cycles, the uniaxial compressive strength drops rapidly after reaching the peak point on the stress–strain curves, and the rock specimens exhibit overall failure, with evident brittle characteristics. In comparison, for rock specimens subjected to F-T cycles at extreme temperatures, their uniaxial compressive strength separately declines by 26% (15 cycles) and 33% (25 cycles). Additionally, failure of the rock specimens is mainly manifest as surface spallation, without exhibiting significant brittle characteristics. When reaching the peak point on the stress–strain curves, the curves tend

to descend, while the rock specimens still retain a certain bearing capacity.

Rock specimens subjected to 0, 15, and 25 F-T cycles differ significantly in terms of the micro-fracture compaction stage. The difference lies in that in the initial stage of stress–strain curves, the superficial layer of rock specimens subjected to F-T cycles shows a large compressible volume. This is mainly because the initial micro-fractures in the rock expand and propagate due to F-T cycling, so the stress needed to induce the same strain decreases. As a result, the rock specimens are extremely sensitive to the stress on a small scale and are easily compressed. This is particularly true for rock specimens undergoing 25 F-T cycles, in which micro-fractures expand to a large volume because of the large temperature difference and large number of F-T cycles. Therefore, the rock specimens are not only extremely easily deformed under stress, but also exhibit a greater strain in the micro-fracture compaction stage, corresponding to a strain rate of 0.7%. After the expansion of micro-fractures induced by F-T cycles has been compressed, the rock enters the elastic deformation stage. In the stage, the micro-fractures in the rock are further closed, which is explained as the further closure of deep micro-fractures in the rock specimens. From this stage onwards, the rock specimens undergo deformation consistent with that of natural rock specimens.

Fig. 4 Stress–strain curves of rock specimens subjected to different numbers of F-T cycles in the uniaxial compression tests



When reaching the peak, rock specimens not subject to F-T cycling remain intact; because a lot of energy is accumulated in the rock specimens, the rock suffers brittle failure under extremely high stress, while the stress drops abruptly in the post-peak stage. For rock specimens experiencing F-T cycles, because of the presence of micro-fractures formed due to F-T cycles in the surface and the coalescence of the surface micro-fractures with growing stress, the surface is spalled after reaching the peak strength. Figure 5 displays the failure modes of rock specimens after different numbers of F-T cycles. Figure 5(a) shows a rock specimen not subjected to F-T cycling: this is fragmented when reaching the peak stress, accompanied by a loud sound, and the fragments are ejected therefrom. Figures 5(b) and (c) separately demonstrate rock specimens experiencing 15 and 25 F-T cycles. The failure zones of rock specimens subjected to F-T cycles are in the surface layer. When reaching the peak point, surface cracks of the rock specimens coalesce and the corresponding location is exfoliated. Therefore, under these F-T conditions, the main difference of rock specimens is generally shown in the micro-fracture compaction stage, as evinced by the propagation and expansion of surface micro-fractures in rock specimens. After the rock specimens enter the elastic deformation stage, when micro-fractures formed due to expansion of the surface layer of specimens are

completely compacted, different groups of rock specimens exhibit similar behavior in terms of the stress–strain curves. This stage can be regarded as a compression of deep micro-fractures in rock specimens, which mainly depends on the internal structural planes in the rock and the development of micro-fractures, while is independent of the F-T conditions.

To estimate the influences of F-T cycles on the mechanical behavior and damage to basalt, it is necessary to perform further detailed analysis on the failure modes of compressed basalt. Thorough exploration of the causes reveals that in, the cyclic F-T process, the F-T effect constantly influences the rock and gradually exacerbates the damage. With the gradual increment of the number of F-T cycles, tiny cracks begin to develop in near-surface area of the rock specimens. The F-T effect on rock specimens is firstly shown on the surface layer and it causes fatigue damage to the surface material, which is gradually exfoliated.

Water fills the pores in the superficial layer of water-saturated rock specimens under the water-wedge effect. Only a small portion of water molecules can permeate to depth in rock specimens. When placing rock specimens in the environment of $-80\text{ }^{\circ}\text{C}$, water in the superficial layer rapidly releases heat due to the extremely large temperature gradient, so the temperature drops abruptly and water freezes rapidly. In the process, the superficial layer of rock specimens

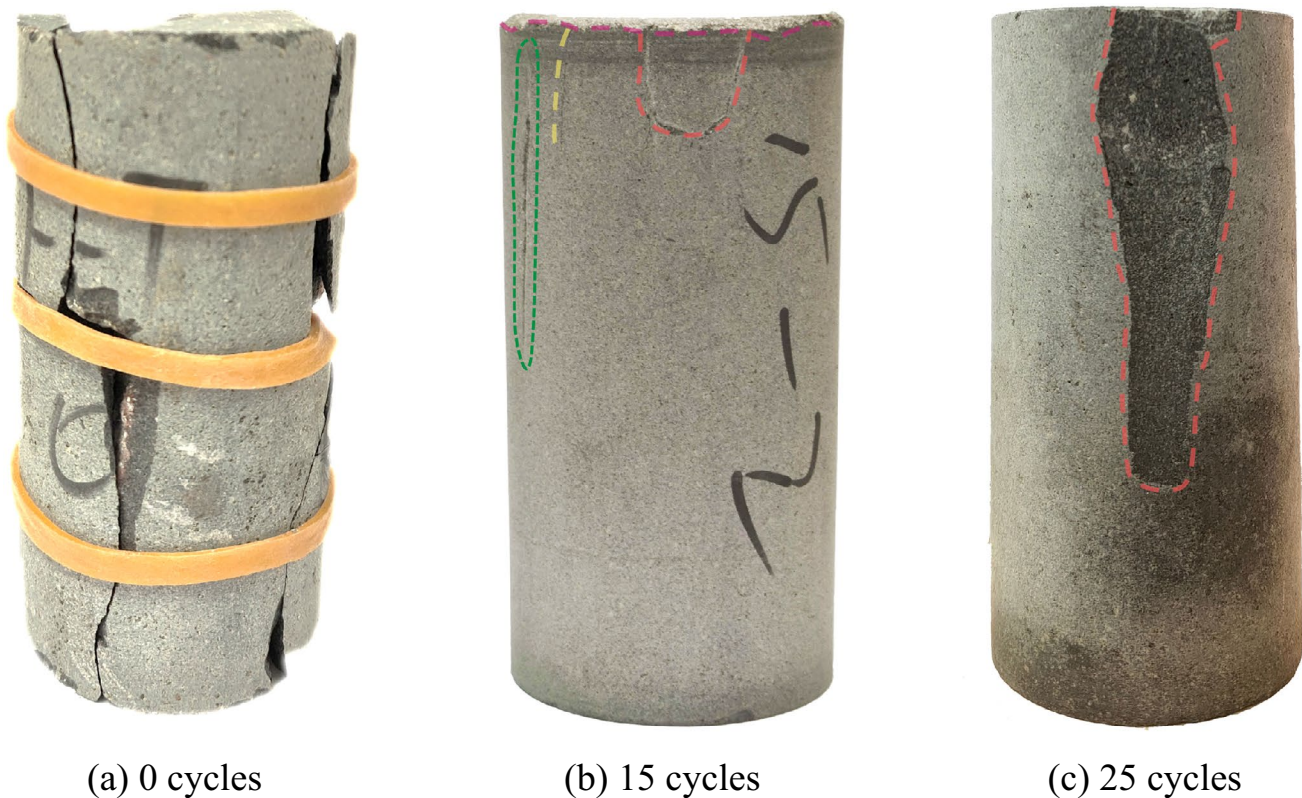


Fig. 5 Failure modes of rock specimens experiencing different numbers of F-T cycles

is subjected to frost heave, so the bonding force reduces and micro-fractures expand. Thereafter, the ice in the superficial layer gradually thaws and water fills the pores in the superficial layer again when placing the rock specimens in an environment heated to 100 °C. Then, the frost heave force is generated in each F-T cycle, that is, when the water freezes. As the F-T cycle is repeated, the superficial layer of the rock continuously suffers fatigue damage induced by F-T cycling.

When applying the compressive load to the rock specimens, the superficial layer suffering F-T damage is more easily exfoliated, thus significantly reducing the overall strength of rock specimens. Such a decrease in the strength is closely related to the F-T damage. Damage within the superficial layer impairs the integrity of rock specimens, substantially weakening the bearing capacity and greatly affecting the mechanical behaviors of rock specimens.

AE monitoring results

The loading of specimens is accompanied by the initiation and propagation of micro-fractures. The occurrence of cracks is accompanied by the emission of signals. Therefore, AE is commonly employed to analyze the failure mechanism of rocks. Figure 6 displays changes in the stress, AE count, cumulative AE count of basalt specimens subjected to different numbers of F-T cycles with time. Figures 6(a), (b), and (c) separately show rock specimens subjected to 0, 15, and 25 F-T cycles.

In the compaction stage, all rock specimens emit a few AE signals. This is because under the external force, the micro-pores and micro-fractures in rock specimens are compressed and elastic strain energy is released, thus generating a few AE signals. Meanwhile, the cumulative AE count also grows. More frequent AE counts are collected from rock specimens subjected to F-T cycles in the initial period of the micro-fracture compression stage, and there are more AE events. These micro-scale responses fundamentally shape the distinct accumulation patterns observed at macroscopic scale. The evolution pattern of accumulated AE counts clearly demonstrates the influence of freeze-thaw cycles on rock damage behavior. For intact specimens, the accumulated AE counts show near-linear growth during uniaxial compression, reflecting the characteristic progressive damage process of sound rock, with only an abrupt surge occurring at the final failure stage corresponding to macroscopic crack coalescence. In contrast, freeze-thaw treated specimens exhibit distinctly different two-phase abrupt characteristics. During initial loading, a sudden increase in AE counts occurs due to the closure of freeze-thaw induced microcracks. Subsequently, when the stress

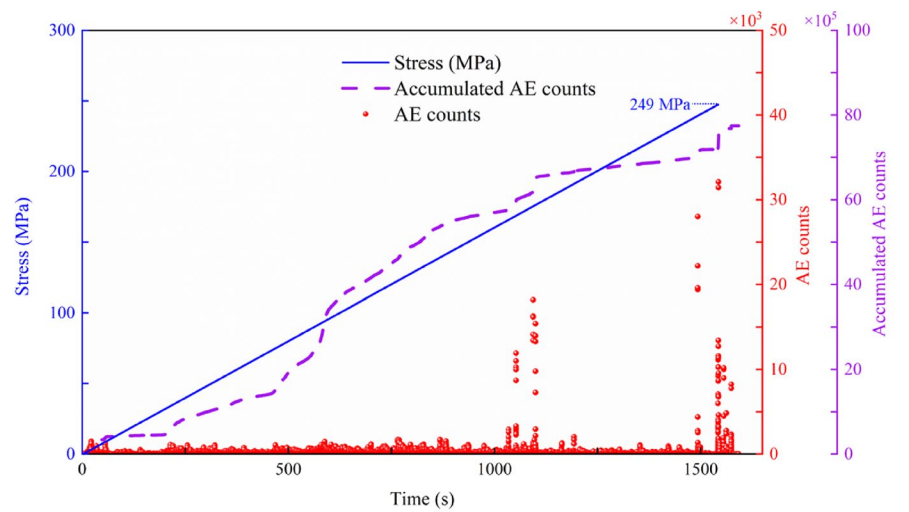
exceeds approximately 80% of peak strength, another count surge is triggered by surface spalling effects.

As the loading process continues, the slope of the cumulative AE count curves of specimens under F-T cycles is approximated to zero. This means that fractures in the specimens are closed under compaction and there are few new AE signals released. As loading continues, the AE count begins to enlarge before reaching the peak stress, suggesting that new cracks are generated in the specimens until the peak stress is reached, when macroscopic failure occurs.

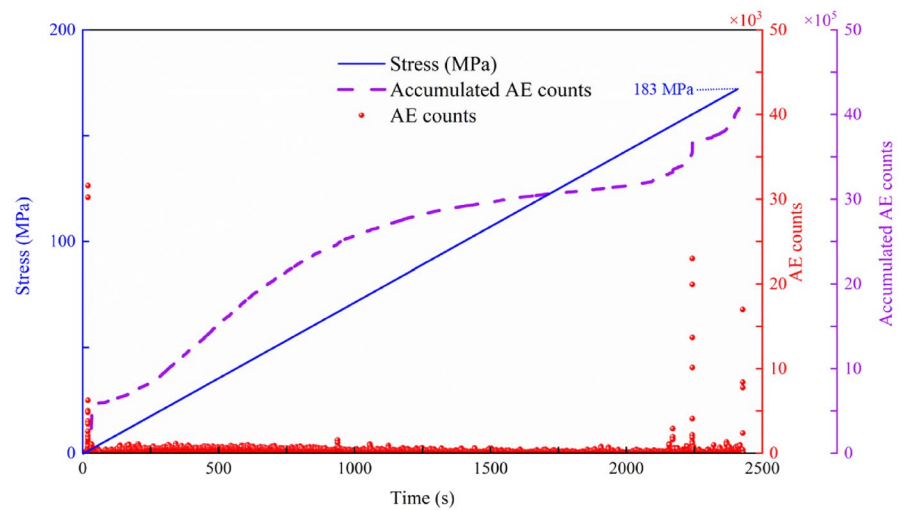
AE signals are generated continuously in the whole loading process of rock specimens not subjected to F-T cycles. The AE positioning technology can well characterize the progressive damage and failure characteristics in the rock. Figure 7 displays the spatio-temporal evolution laws and the corresponding macroscopic damage to rock specimens experiencing different numbers of F-T cycles at crack location points in 20%, 60%, and 80% of the total time. The scatters represent AE events, the size and color of which represent the energy difference between events. Figures 7(a), (b), and (c) separately show rock specimens subjected to 0, 15, and 25 F-T cycles. According to the distribution and energy of scatters in Figs. 7(b) and (c), the light-blue area delineates the main area subject to F-T damage. The energy distribution of AE signals provides critical insights into the damage evolution of basalt specimens under freeze-thaw cycles. For untreated specimens, energy release occurs uniformly throughout loading, reflecting progressive microcrack development. In contrast, F-T-treated specimens exhibit bimodal energy release patterns: initial low-energy signals dominate during microcrack closure in superficial layers, while high-energy events emerge later near peak stress, coinciding with macro-fracture formation. Notably, the spatial clustering of high-energy events around pre-existing low-energy zones highlights stress concentration areas weakened by F-T damage. Specimens with more F-T cycles generate significantly more low-energy events early in loading, indicating accelerated damage accumulation. This transition in energy release patterns—from gradual to abrupt—demonstrates that F-T cycles deteriorate rock integrity, diminish energy storage capacity, and ultimately induce premature failure. The areas with F-T damage after 15 and 25 F-T cycles separately account for 31.6% and 46.7% of the total, and they are within the ranges of radial distances of 7.9 mm and 11.7 mm from the surface.

During compression of rock specimens not subject to F-T cycles, cracks show a specific spatio-temporal evolution. The number of scattered data points exhibits uniform growth with time; however, when the rock specimens are about to fail, the number of scattered data points increases abruptly. At this moment, the areas where events are concentrated

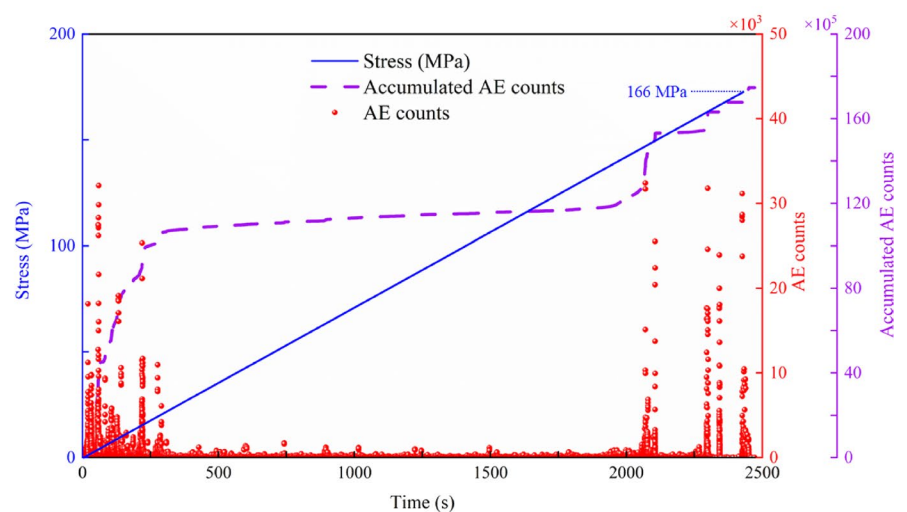
Fig. 6 Distribution of AE counts of basalt specimens subjected to different numbers of F-T cycles



(a) 0 cycles



(b) 15 cycles



(c) 25 cycles

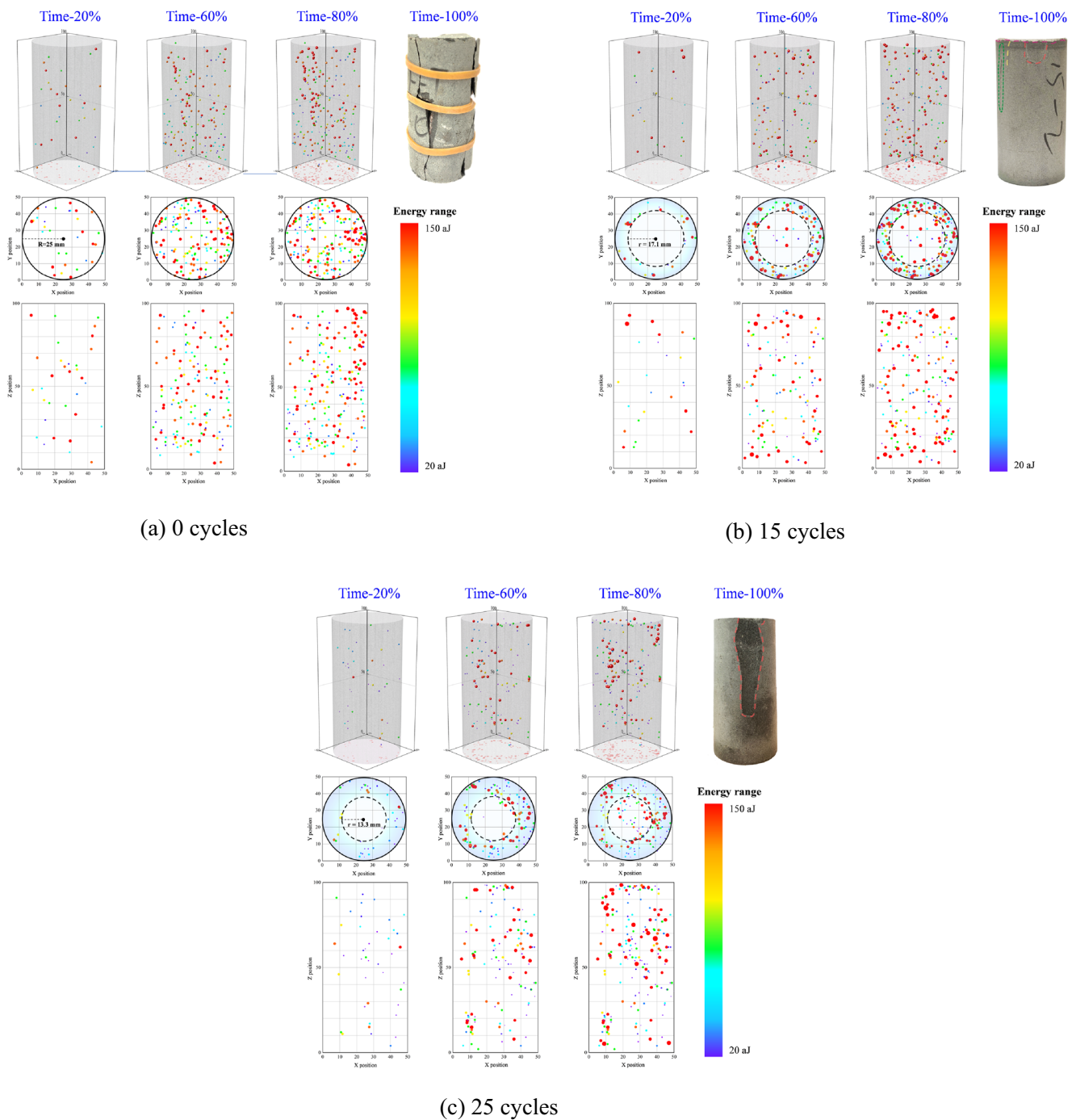


Fig. 7 The relationship between crack development and macroscopic damage to basalt specimens experiencing different numbers of F-T cycles

match the areas of failure. This phenomenon fully indicates that in the whole compression process, mineral particles inside rock specimens are gradually damaged due to the continuous action of compressive stress, thus continuously releasing lots of energy signals. For rock specimens subjected to F-T cycles, cracks developed in the initial stage of compression are mainly concentrated in the superficial layer and correspondingly, low energy is released. Over

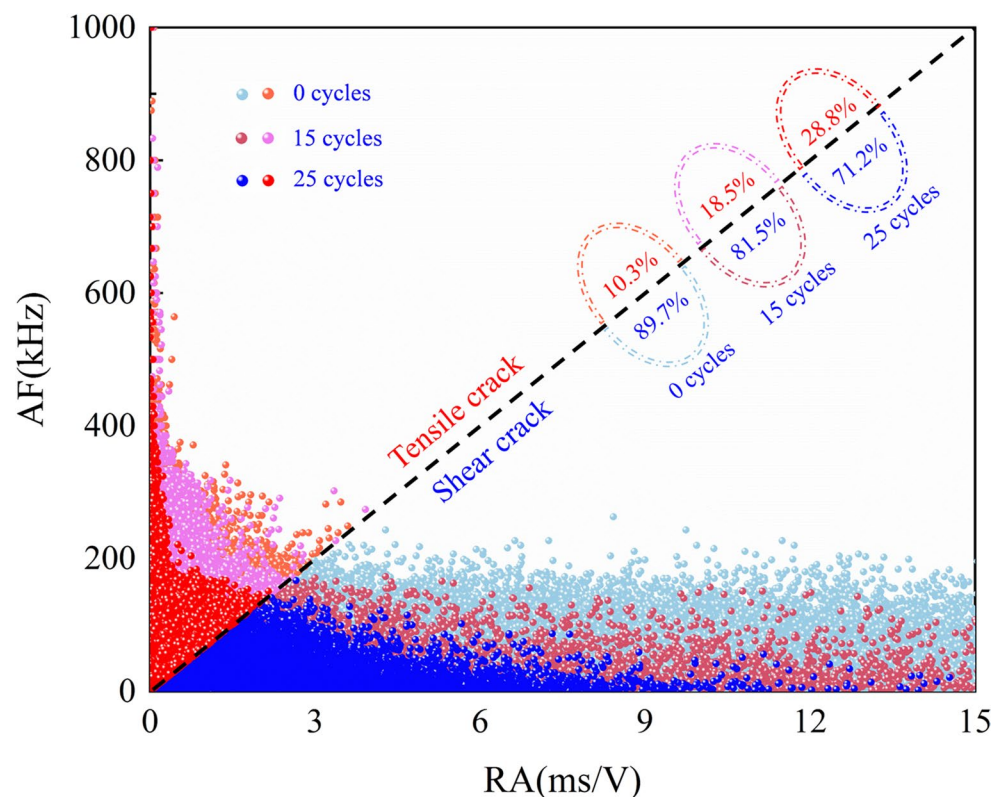
time, more high-energy signals are generated in rock specimens and this continues until the specimens are on the verge of failure, when many signals are emitted. This means that the damage and cracks formed due to the F-T effect in the superficial layer of rock specimens subjected to F-T cycles further grow and develop as compression progresses in the initial stage. Meanwhile, low-energy signals different from those in conventional rock specimens are generated.

It is worth noting that high-energy signals emitted in the late stage of compression are almost all located around the low-energy signals. This suggests a low strength due to the significant influence of FT-induced damage, so macro-cracks occur preferentially at this location upon failure of the specimens. Further comparison of rock specimens subjected to 15 and 25 F-T cycles reveals that the two exhibit significant differences: the more numerous the F-T cycles, the larger the number of low-energy events generated in the initial stage of compression. This strongly suggests that the number of F-T cycles exerts a significant influence on the integrity of internal structures and the mechanical response characteristics of rock specimens to the compression. Increasing the number of F-T cycles exacerbates the degree of damage to the interior of the rock specimens and allows the crack initiation and propagation to occur earlier in the subsequent compression process. This thus changes the overall mechanical behaviors and failure modes of rock specimens and allows specimens to fail under low stress. Additionally, the failure mode and process also show apparent differences under different numbers of F-T cycles.

Analysis of the relationship between the ratio of rise time to amplitude (RA) and the average frequency (AF) is one of the more commonly used methods for analyzing AE signals (Liu et al. 2019; Kong et al. 2025). The Japan Building Materials Federation was the first to propose use of the ratio of AF to RA as the classification criterion for

distinguishing shear cracks from tensile cracks. A low RA while a high AF mean the occurrence of tensile cracks in rocks; a high RA while a low AF suggest the occurrence of shear cracks in rocks (Ohno and Ohtsu 2010). Figure 8 displays the RA-AF scatter density distribution of rock specimens subjected to different numbers of F-T cycles under uniaxial loading. The diagonal is selected as a boundary for tensile-shear failure. That is, the area above the boundary represents tensile failure, while that below the boundary represents shear failure. In the failure of rock specimens under uniaxial compression, the proportion of shear failure changes from 89.7 to 81.5%, and finally to 71.2%. For natural rock specimens, most failure under compression of the external load is shear failure, which means that, as the number of F-T cycles augments, tensile failure gradually develops in the specimens. The phenomenon proves that the water-ice phase change in the F-T cycles causes repeated frost heaving of micro-fractures in the superficial layer of the rock, and the resulting frost heave force exerts a significant influence at the end of the cracks, thus inducing tensile failure of micro-fractures. As the number of micro-fractures increases, the number of micro-fractures shown as tensile failure induced by the frost heave force continues to grow, so that the rock structure loses its original integrity and stability. Water freezes to ice and expands. For pre-existing micro-fractures in the superficial layer of the rock, the frost heave force acts as a wedge to apply tension at the

Fig. 8 Distribution of tensile and shear cracks in basalt subjected to different numbers of F-T cycles



end of cracks, so that the cracks propagate and extend along the action direction. Each F-T cycle aggravates the failure effect. Macroscopically, the accumulated failure is manifest by severe deterioration of overall mechanical behaviors and the gradual enlargement of the proportion of tensile failure in the rock.

As the loading continues, the internal structure of the rock is further weakened in the plastic stage, in which the interaction of cracks becomes more complex. Due to development of tensile cracks in the previous stages, the integrity of the rock has been impaired, so the rock shows more of the characteristics of tensile failure upon final failure. The proportion of tensile failures in the rock specimens increases from 10.3 to 18.5% (15 cycles), and finally to 28.8% (25 cycles).

SEM results

Rocks are mainly constituted by mineral particles and their interior can be divided into two parts: particles and pores (Li et al. 2023). With regard to basalt specimens selected in the research, pores therein are all filled with water in each F-T cycle. When the ambient temperature drops abruptly to -80°C , the pore water freezes to ice and expands. The formation of ice increases the pressure inside the pores in the rock. Particularly in tiny pores in the rock, the expanding ice exerts a mechanical pressure inside the pores. The pressure is likely to induce propagation of micro-fractures in the rock, thus impairing the overall structure and strength of the rock.

The weak planes in the crystals of the rock are mainly composed of particle boundaries of the rock. The failure induced by intergranular (IG) fracture generally initiates from, and extend along, grain boundaries, belonging to ductile-brittle failure; failure induced by transgranular (TG) fracture generally goes through whole grains, typifying brittle failure (Kranz 1983; Zhang and Zhao 2014; Wang et al. 2015). Although the two failure modes both damage the rock, they differ in terms of the process and the internal mechanism. To study the influence of F-T cycles on microstructures of basalt specimens, SEM was adopted to observe microstructures of damaged rock specimens experiencing and not experiencing F-T cycles (Fig. 9). Figures 9(a), (b), and (c) separately exhibit rock specimens subjected to 0, 15, and 25 F-T cycles. Before SEM, rock specimens were fully washed with distilled water, followed by soaking in pure ethanol solution for 6 h and then washing with distilled water again. This operation was repeated three times. After treatment, the rock specimens were dried in an electric drying oven, followed by gold spraying to enhance the electrical conductivity and improve the clearness of SEM images.

Comparison of rock specimens subjected to different numbers of F-T cycles indicates that the rock specimens experiencing more F-T cycles show clean fracture planes. New fractures have sharp and significant corners, with high contrast in the SEM images. To clarify the results, those images with the maximum magnification times in each group were binarized and then images in the three groups were compared with the same threshold. If the intensity exceeds the threshold, it is determined that there are signals, corresponding to the bright part in the image; if the intensity is lower than the threshold, then it is considered that there are no signals, corresponding to the dark part in the image (Zhu et al. 2020). Through the above binarization processing and analysis, Fig. 10 shows that the rock specimen not subjected to F-T cycling contains fewer new fractures at failure. The continuous bright areas in the images can be regarded as the new fractures inside the rock induced by the F-T cycles. This provides an intuitive and powerful basis for exploring the influence of F-T cycles on microstructures of the rock.

According to the binarization processing results, fracture characteristics of rock specimens with different numbers of F-T cycles are marked in Fig. 10 by comparing with areas with high contrast in the SEM images. The red area indicates fractures of mineral particles under stress; blue area suggests smooth failure of mineral particles. Areas with high contrast in the SEM images mainly undergo IG fracture. For basalt specimens not affected by F-T cycles, SEM observation shows that particles are tightly contacted and uniformly distributed. These specimens are fragmented with TG fracture characteristics, and cracks go through mineral particles to form macro-cracks, thus significantly lowering the strength of the rock. The fracture mode occurs because the internal structure of the particles is intact, so it can bear a high compressive stress and has a high strength. The result also confirms the theory of splitting failure. On the contrary, basalt specimens subjected to F-T cycles show more obvious IG failure characteristics in the images after compression tests. The failure mode occurs because of the increase in the porosity and the propagation of micro-fractures caused by the huge release of heat and freezing of pore water due to the sudden temperature drop under F-T cycles under the extremely large temperature difference. Due to the increment of the porosity, the internal structure of the rock becomes loose and the freeze heaving and thawing of water in the pores are accompanied by the propagation and development of associated fractures. Therefore, for rock specimens subjected to cryogenic F-T cycles, the cracks propagate along the new associated fractures under the compressive stress,

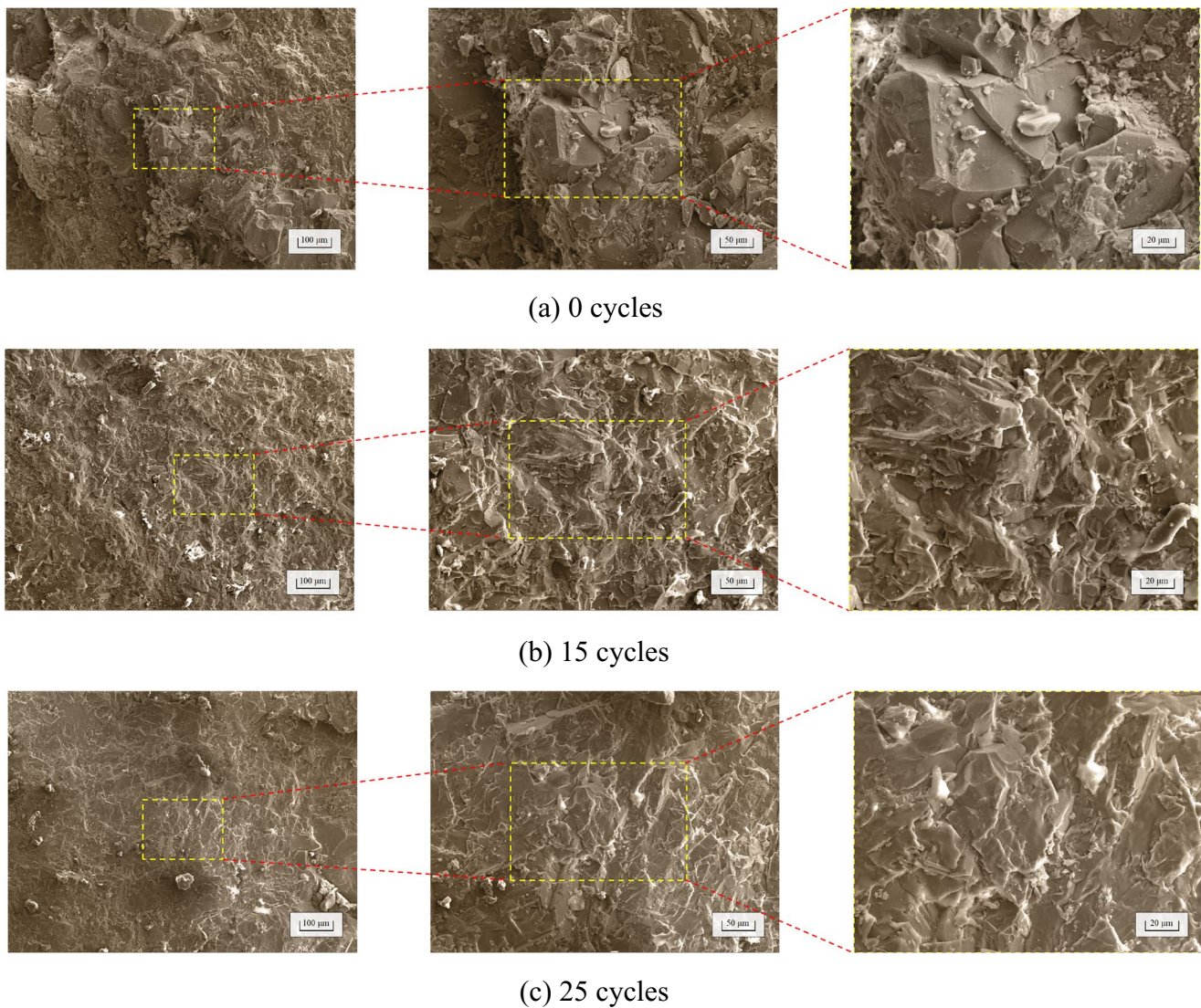


Fig. 9 SEM analysis of failure morphology of rock specimens under different conditions

which causes surface spalling of the rock and therefore significantly reduces the rock strength.

SEM results confirm that for basalt specimens damaged in the compression test, rock specimens not subjected to F-T cycles can better resist the high compressive stress due to the relatively intact mineral particles and internal structure. Macro-cracks develop and the specimens fail only when the stress reaches the maximum level that can be borne by crystals. However, in basalt specimens subjected to F-T cycles with an extremely large temperature difference, pore water pressure forces the occurrence of more micro-fractures in the intergranular region inside the rock. When applying the compressive stress, the growth and development of these micro-fractures not only exacerbate the surface spalling of the rock but also weaken the rock.

Discussion

Frequency and energy

The peak frequency and amplitude in AE characteristics are important parameters for studying rock failure. Therein, high-frequency signals correspond to small cracks, while low-frequency ones correspond to large cracks (Carpinteri et al. 2016; Smith et al. 2019; Wang et al. 2024). Figure 11 illustrates the relationships of the amplitude and peak frequency at various points collected by the AE apparatus in the loading process of rock specimens subjected to different numbers of F-T cycles with time. The peak frequency is divided into three sections: namely, a low-frequency section (0~200 kHz), a medium-frequency section (200~400

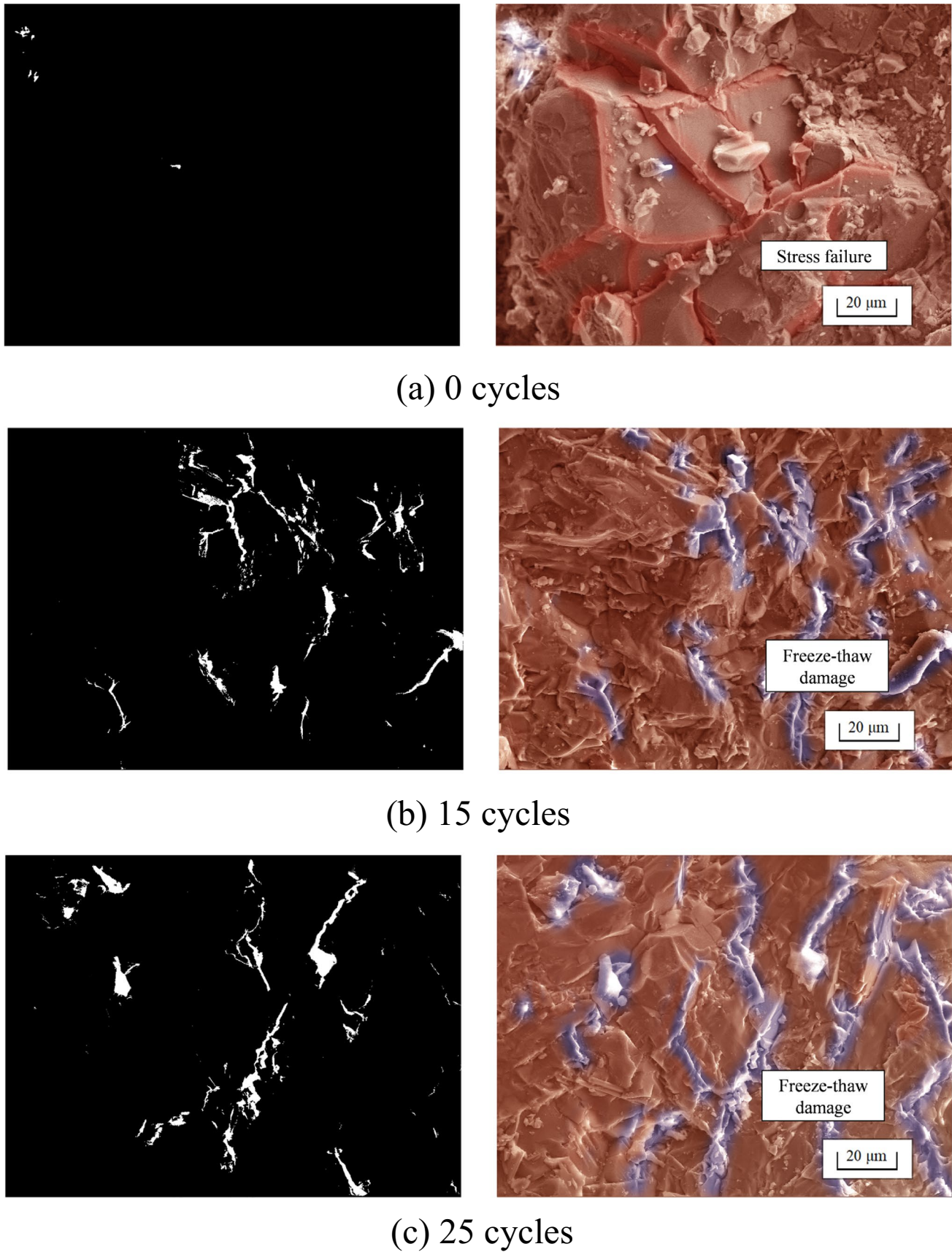


Fig. 10 Difference in the binarization processing results of stress and F-T damage

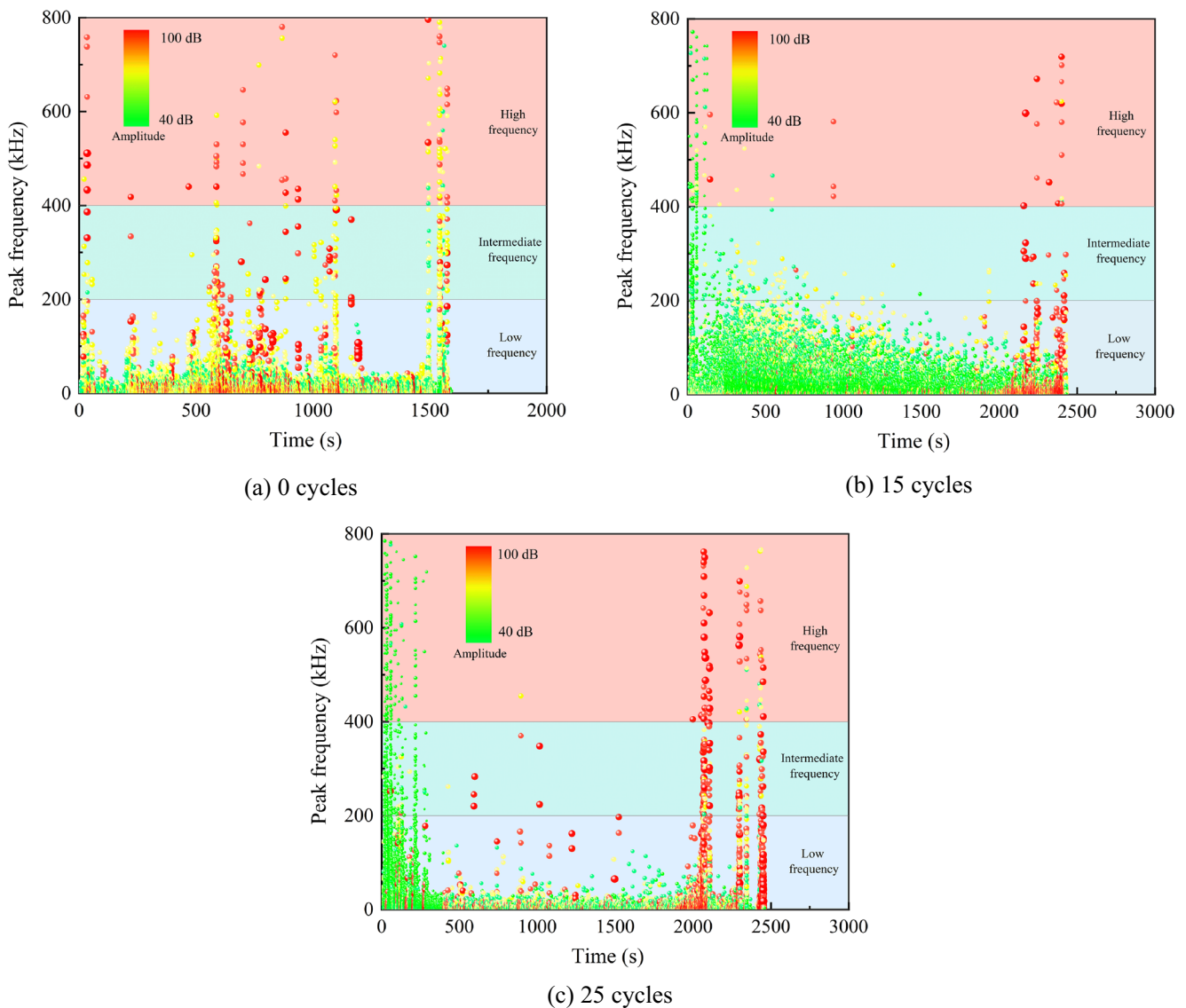


Fig. 11 Peak frequency-amplitude distribution of rock specimens subjected to different F-T cycles

kHz), and a high-frequency section (400~800 kHz). The amplitude of each point is mapped by the color and size. In the three groups of rock specimens, low-frequency signals increase near the peak stress, which suggests that cracks gradually propagate and coalesce to form macro-fractures. The peak frequency and amplitude are distributed uniformly in the whole loading process of rock specimens not experiencing F-T cycles, and they show a step-by-step increasing trend with time. This proves that without F-T cycles, the internal structure of the rock is relatively stable and uniform. In addition, the crack initiation, propagation, and interaction in the loading process proceed relatively smoothly. Initially, only a few micro-fractures occur, then the micro-fractures gradually increase and slowly develop with increasing load, and finally they form macro-fractures. In the process, their

mechanical responses exhibit certain progressive variation in the frequency and amplitude, reflecting a conventional fracture-development history of rock under load.

For rock specimens subjected to cryogenic F-T cycles, the high-frequency low-amplitude signals are concentrated in the initial loading stage, that is, the crack compaction stage. These high-frequency low-amplitude signals indicate the rapid generation of many low-energy new cracks, and with the increasing number of F-T cycles, the longer the time for generation of such signals. The more numerous the cryogenic F-T cycles, the more severe the damage accumulated within the rock. More micro-fractures and more severe damage to the internal structure render the rock more likely to emit signals throughout the loading process. Due to the complexity and diversity of damage, the high-frequency

low-amplitude signals persist for a longer time. Combined with points of AE event, the areas that emit signals at the moment are mainly found in the superficial layer of the rock specimens, which is regarded as the main areas of F-T damage, corresponding to the high-frequency and low-amplitude characteristics of signals. This indicates that the damage to the mechanical behaviors of the rock due to F-T cycles is accumulated, which not only changes the fracture mode of the rock but also prolongs the unstable state in the loading process. In terms of the AE characteristics, this is shown as the extension of the time for generating specific signals and also indicates that the overall strength and stability of the rock are deteriorated significantly with the increasing number of F-T cycles. In engineering applications, the rock structures subjected to multiple F-T cycles are more prone to failure and the warning time before failure is longer and the process is more complex.

Microscopic evolution of fractures

The rock specimens not subjected to F-T cycles exhibit typical brittle failure. Real-time AE monitoring shows that such brittle failure is a process in which the energy is gradually accumulated, manifest in the TG failure of mineral particles (microscopic analysis confirms this). In the state without F-T cycles, the structural integrity of mineral particles themselves is dominated in resistance to failure due to the external force. When applying the external load to the rock specimens, the stress is transferred to within the mineral particles. Due to the possible presence of local stress concentration points in mineral particles or strength inhomogeneity, fractures initiate in the particles in priority. Once these micro-fractures are formed, they rapidly pass through mineral particles and propagate under stress, finally causing TG failure and thus brittle fracture of the whole rock specimen.

In the process of cryogenic F-T cycles, due to the ambient temperature drops suddenly to -80°C , the abrupt change of temperature causes the pore water in the rock to release a huge amount of heat and freeze rapidly. The rapid expansion of the ice generates an extremely large frost heave force, so that the pre-existing micro-fractures in the superficial layer of rock specimens to extend and propagate. Whereas, the structure of single mineral particles remains quasi-intact, with a high strength remaining, even after applying the compressive stress. This suggests that the mineral particles themselves are capable of resisting F-T damage. The new fractures observed from SEM images are induced by F-T cycling, and they show the significant characteristics of IG failure at the microscopic scale. AE monitoring signals have high energy while a low amplitude and are microscopically manifest as IG failure. The more the F-T cycles, the larger the number of new fractures induced by F-T cycles, which

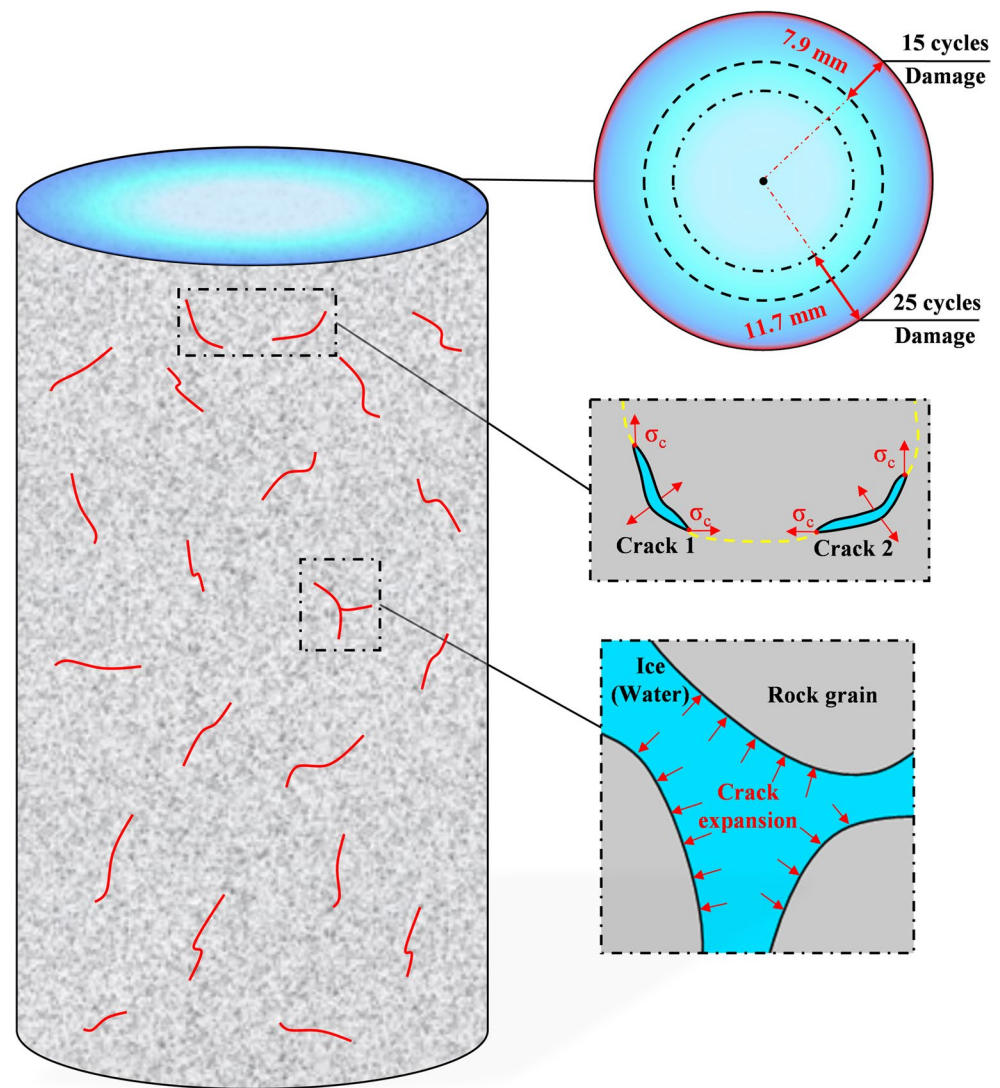
gradually weaken the binding force at the grain boundaries of mineral particles. As a result, grain boundaries are more likely to become weak areas prone to crack initiation and propagation, and the proportion of tensile failure in the rock also increases step-by-step. The new fractures induced by F-T cycling microscopically correspond to IG failure between particles, thus causing more tensile failure in the rock specimens.

Damage characteristics of rock specimens

The mechanical strength of basalt specimens tends to reduce with the increasing number of F-T cycles under low-temperature conditions. In the cyclic F-T process, micro-fractures and associated fractures develop in the rock specimens. When the micro-fractures develop to a certain extent, they form larger fracture networks. These fracture networks interweave to form complex channels and cavities in the rock. Under such conditions, P waves not only face more scattering and reflection during propagation, but also their energy is absorbed and attenuated due to air or other media in the cracks when they pass through cracks. As a result, the P-wave velocity further declines, and such a decreasing trend becomes increasingly obvious with the further development of the fracture network. Considering that basalt is classified as a low-permeability rock and the F-T damage is gradually transferred from the superficial layer to the deep layer in the F-T process, F-T damage is mainly concentrated in the superficial layer. In comparison, the internal structure of the rock specimens remains relatively stable, and the abnormal mechanical behaviors are also mainly found in the superficial layer. This is confirmed by both the loci and intensity of AE events and the different failure modes of rock specimens.

Figure 12 displays the evolution of fracture networks in basalt subjected to cryogenic F-T cycling. In Fig. 12, σ_c signifies the stress concentration at crack tips generated by localized stresses from the 9% volumetric expansion during water-ice phase transition. Throughout this F-T process, pore water in the superficial layer of rock specimens freezes and thaws repeatedly, and the resulting frost heave generates fatigue damage in the superficial layer, allowing micro-fractures to propagate continuously and coalesce. When the compressive load is applied, the superficial layer of rock specimens is damaged, cracked, and begins to be spalled under low stress, whereas, the interior is only slightly affected by F-T cycling and retains a certain bearing capacity. Compared with rock specimens not undergoing F-T cycling, those subjected to F-T cycles show significant difference in the micro-fracture compaction stage. Micro-fractures generated in the superficial layer of rock specimens due to F-T cycles continuously propagate under the

Fig. 12 Evolution of fracture networks in basalt during cryogenic F-T cycles



compressive stress, while the interior close to the center is hardly affected by F-T cycles. This indicates that damage caused by F-T cycling to the rock does not occur uniformly, but gradually develops from the superficial layer inwards. The areas with F-T damage after 15 and 25 F-T cycles separately account for 31.6% and 46.7% of the total, and they are within the ranges of radial distances of 7.9 mm and 11.7 mm from the surface.

Due to the extreme freezing temperature and the larger temperature difference, when transferring the basalt specimens from the environment at 100 °C to that at −80 °C, water releases a huge amount of heat energy and rapidly freezes. Compared with those F-T cycles with a smaller temperature difference, the water freezes more rapidly in the above case; because water freezes almost instantaneously at an extremely low temperature, pore water in the rock is transformed rapidly into ice, which gives rise to the

significant internal stress concentration, induces the initiation and propagation of micro-fractures, and thus damages the rock. As a result, the rock is more easily damaged under the external load. This has also been verified by the analysis of mechanical behaviors and SEM analysis. Such a non-uniform transition of failure modes provides a theoretical basis for understanding the damage mechanism of rock under long-term F-T action in a cold region. It should be emphasized that the failure modes and energy evolution characteristics revealed in this study are primarily applicable to hard rocks with certain porosity in cold regions, where damage is predominantly governed by volumetric expansion from water-ice phase transition. For temperature-sensitive soft rocks or highly permeable rock masses, the freeze-thaw induced strength degradation and damage extent could be more pronounced due to their heightened thermal sensitivity or enhanced moisture migration.

Conclusion

Uniaxial loading tests were conducted on basalt specimens subjected to cryogenic F-T cycles ($-80\text{ }^{\circ}\text{C} \sim 100\text{ }^{\circ}\text{C}$). AE and SEM were utilized to monitor and analyze the damage evolution process in the specimens. The test results provide basic data for research into the mechanical behaviors of rocks with F-T damage and at the same time, provide theoretical guidance for building construction and slope design and construction in alpine regions. The following main conclusions are drawn:

- 1) The P-wave velocity decreases with the increasing number of F-T cycles. It reduces by 12% from 5834 m/s (0 cycle) to 5113 m/s (25 cycles). As the number of F-T cycles augments, the crack development and propagation cause the P-wave velocity to decrease;
- 2) F-T cycles change the brittle failure of rock specimens under peak stress to the coalescence and spalling of surface cracks. After 15 and 25 F-T cycles, the uniaxial compressive strength of the rock specimens separately reduces by 26% and 33%. The initial micro-fractures expand and propagate due to F-T cycles, so that the stress required to reach the same strain decreases in the initial loading stage. The frozen-thawed rock specimens are more sensitive to the stress on a small scale;
- 3) AE results show that the areas with F-T damage separately account for 31.6% and 46.7% of those rock specimens subjected to 15 and 25 F-T cycles, and they are separately in the ranges with radial distances of 7.9 mm (15 cycles) and 11.7 mm (25 cycles) from the surface. Through F-T cycling, more high-frequency low-energy scatters are generated in the early stage of compression, and the proportion of tensile failure in rock specimens enlarges from 10.3 to 18.5% (15 cycles), and finally to 28.8% (25 cycles). Macroscopically, the accumulated damage is finally shown as severe deterioration of overall mechanical behaviors of the rock;
- 4) For rock specimens not subjected to F-T cycles, their fracture planes show significant characteristics of TG fracture at a microscopic scale. The crystal texture in the mineral particles is relatively intact and shows strong compression resistance. Rock specimens subjected to F-T cycles exhibit more prominent IG failure characteristics. Micro-fractures begin to grow and develop in priority along new fractures formed due to F-T cycles, thus weakening the strength of such rock specimens.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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